医院信息化网络改造设计

一、用户需求

1. 医院规模

- 三级甲等医院
- 开放床位：800 张

2. 网络分区

- 医疗内网：承载 HIS、PACS、LIS 等核心医疗业务系统
- 办公外网：行政、财务、互联网访问等办公业务
- 设备专网：医疗设备、物联网终端、监护设备等专用网络

3. 终端数量

- 医生工作站：300 台
- 护士工作站：200 台
- 行政办公电脑：150 台
- 移动医疗终端：100 台

4. 安全要求

- 内外网**物理隔离**
- 医疗数据**加密传输**
- 上网行为**审计与管理**

5. 可靠性要求

- 核心设备**双机热备**
- 主干链路**冗余备份**
- 满足 **7×24 小时**不间断运行

二、设计要求

1. 撰写网络改造**可行性分析报告**
2. 设计三**网分离**的网络架构
3. 规划**无线网络覆盖**（病房、门诊区域）
4. 设计**网络安全防护体系**
5. 设备选型与**配置清单**
6. 配置核心交换机 **VRRP、链路聚合**

虚拟环境 Cisco Packet Tracer 7.0.0

1 ip分配

网络分区	用途	VLAN ID	设备数量	网段	网关
医疗内网	300台医生工作站	10	300	10.10.10.0/23	10.10.11.254
	200台护士工作站	20	200	10.10.20.0/24	10.10.20.254
	医疗内网服务器，完成HIS，PACS，LIS	30	1 扩展冗余12	10.10.30.0/28	10.10.30.14

网络分区	用途	VLAN ID	设备数量	网段	网关
办公外网	150台行政办公电脑	210	150	10.10.210.0/24	10.10.210.254
	外网路由器	220	1 扩展冗余12	10.10.220.0/28	10.10.220.14
	财务行政服务器	230	1 扩展冗余12	10.10.230.0/28	10.10.230.14
	客户病房无线网络覆盖	240	4094	10.20.0.0/20	10.20.15.254
设备专网	100台医疗移动终端	110	100	10.10.110.0/25	10.10.110.126
	设备信息服务器	120	1 扩展冗余12	10.10.120.0/28	10.10.120.14
管理网络	网络设备管理	40		10.10.40.0/24	10.10.40.254
	无线AP	50	85	10.10.50.0/24	10.10.50.254
	设备AP	60	85	10.10.60.0/24	10.10.60.254

本设计采用私有IPv4地址空间 `10.0.0.0/8`，并选用 `10.10.0.0/16` 作为核心地址块，同时为无线访客网络单独划分 `10.20.0.0/20`，以避免与内网地址重叠。所有VLAN的划分遵循以下原则：

- 每个VLAN对应一个独立的子网，用于隔离不同业务或管理流量。
- 子网大小根据终端数量、冗余需求和未来扩展进行估算。
- 网关地址通常取子网内最后一个可用IP地址（如 `.254` 或 `.14`），便于记忆和统一管理。
- 管理VLAN（40、50、60）独立于业务VLAN，确保设备管理安全。

以下为各VLAN的详细计算过程：

1.1 医疗内网

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
10	医生工作站	300	/23	512	10.10.10.0	10.10.11.255	10.10.10.1 ~ 10.10.11.254	10.10.11.254	300台终端需至少300个IP，/23提供512个地址，满足当前并留有约212个扩展空间。网关选用最后一个可用地址。
20	护士工作站	200	/24	256	10.10.20.0	10.10.20.255	10.10.20.1 ~ 10.10.20.254	10.10.20.254	200台终端使用/24（256地址），足够并留有56个冗余。

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
30	医疗内网服务器	1 (+12)	/28	16	10.10.30.0	10.10.30.15	10.10.30.1 ~ 10.10.30.14	10.10.30.14	服务器初期仅1台，但考虑到冗余和未来可能增加，采用/28提供14个可用地址（除去网络地址和广播地址），网关为最后一个可用地址.14。

1.2 办公外网

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
210	行政办公电脑	150	/24	256	10.10.210.0	10.10.210.255	10.10.210.1 ~ 10.10.210.254	10.10.210.254	150台终端，/24足够，留有103个扩展。

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
220	外网路由器	1 (+12)	/28	16	10.10.220.0	10.10.220.15	10.10.220.1 ~ 10.10.220.14	10.10.220.14	路由器接口只需少数地址，/28提供14个可用，可容纳未来多出口或备用路由器。
230	财务行政服务器	1 (+12)	/28	16	10.10.230.0	10.10.230.15	10.10.230.1 ~ 10.10.230.14	10.10.230.14	同服务器区，预留扩展。

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
240	客户病房无线网络	4094	/20	4096	10.20.0.0	10.20.15.255	10.20.0.1 ~ 10.20.15.254	10.20.15.254	无线终端数量难以精确估算，但为满足大量移动设备接入，使用/20（4096地址），足够覆盖全院访客。网关同样取最后一个可用地址。

1. 3 设备专网

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
110	医疗移动终端	100	/25	128	10.10.110.0	10.10.110.127	10.10.110.1 ~ 10.10.110.126	10.10.110.126	100台移动终端使用/25提供128地址，满足并留有28个扩展。

VLAN	用途	终端数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
120	设备信息服务器	1 (+12)	/28	16	10.10.120.0	10.10.120.15	10.10.120.1 ~ 10.10.120.14	10.10.120.14	服务器区，/28足够。

1.4 管理网络

管理网络独立于业务网络，用于所有网络设备（交换机、路由器、防火墙、无线控制器等）的带外管理。

VLAN	用途	设备数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
40	网络设备管理	约80	/24	256	10.10.40.0	10.10.40.255	10.10.40.1 ~ 10.10.40.254	10.10.40.254	核心、汇聚、接入、防火墙等设备总数约80台，/24可容纳254个设备，满足并留有大量冗余。

VLAN	用途	设备数量	子网掩码	子网大小	网络地址	广播地址	可用地址范围	网关地址	计算说明
50	办公外网AP管理	85	/24	256	10.10.50.0	10.10.50.255	10.10.50.1 ~ 10.10.50.254	10.10.50.254	80台AP + 4台PoE交换机 + 1台WLC = 85个设备，/24足够。
60	设备专网AP管理	85	/24	256	10.10.60.0	10.10.60.255	10.10.60.1 ~ 10.10.60.254	10.10.60.254	同理，85个设备，/24满足。

1.5 地址分配说明

- **网关位置**：统一采用子网内最后一个可用IP作为网关（如 `/24` 的 `.254`，`/28` 的 `.14`，`/25` 的 `.126`），便于记忆和脚本配置。
- **地址范围预留**：每个子网的第一个可用地址（如 `.1`）通常分配给服务器或关键设备，其余地址通过DHCP动态分配。静态设备（如网络设备管理地址）在对应管理VLAN内连续分配，避免与动态池冲突。
- **扩展性**：所有子网均预留了至少20%的冗余地址，部分子网（如医生工作站/23）预留更多，以适应未来医院信息化发展。

2 网络拓扑设计

2.1 设备清单

设备名称	型号	数量	所属网络	角色	IP地址/管理地址
Core-Int-1 / Core-Int-2	Cisco 3650-24PS	2	医疗内网	核心交换机（VRRP、DHCP、三层网关）	10.10.40.1, 10.10.40.2（管理VLAN 40）
Dist-Int-Hospital-1/2	Cisco 3560-24PS	2	医疗内网	汇聚交换机 护士用 4（**带宽扩展 + 冗余**）+ 14	10.10.40.3 10.10.40.4（管理VLAN 40）
Dist-Int-OPD-1/2	Cisco 3560-24PS	2	医疗内网	汇聚交换机 医生用 4 + 14	10.10.40.5 10.10.40.6（管理VLAN 40）
Acc-Int-Doctor	Cisco 2960-24TT	14	医疗内网	医生工作站接入14 * 22 = 308 1个端口作上行通讯	10.10.40.7-20（管理VLAN 40，每台一个）
Acc-Int-Nurse	Cisco 2960-24TT	10	医疗内网	护士工作站接入 10 * 22 = 207 1个端口作上行通讯	10.10.40.21-30（管理VLAN 40）
Server-Int-Switch	Cisco 2960-24TT	1	医疗内网	服务器接入交换机	10.10.40.32（管理VLAN 40） 密码cisco
Int-Server	Server-PT	1	医疗内网	医疗内网服务器	10.10.30.2（业务VLAN 30）

设备名称	型号	数量	所属网络	角色	IP地址/管理地址
Docter-Computer	PC	1	医疗内网	医生电脑	dhcp 获取 （文件中为 10.10.10.4）
Nurse-Computer	PC	1	医疗内网	护士电脑	dhcp 获取 （文件中为 10.10.20.3）
Core-Ext-1 / Core-Ext-2	Cisco 3650-24PS	2	办公外网	核心交换机（VRRP、DHCP、三层网关）	10.10.40.33, 10.10.40.34（管理VLAN 40）
Dist-Ext-Admin-1/2	Cisco 3560-24PS	2	办公外网	行政楼汇聚交换机	10.10.40.35 10.10.40.36（管理VLAN40）
Acc-Ext-Office	Cisco 2960-24TT	7	办公外网	行政办公接入8*22 = 156	10.10.40.37-44（管理VLAN 40）
Dist-Ext-Wireless-1/2	Cisco 3560-24PS	2	办公外网	无线汇聚交换机 4端口上行，1端口wlc ，4端口ap接入交换机	10.10.40.45 10.10.40.46（管理VLAN 40）
Dist-Ext-WLC-1/2	Cisco WLC-2504	2	办公外网	无线网络控制器	10.10.50.1（管理VLAN 50）10.10.50.86
Acc-Ext-AP-Switch	Cisco 2960-24PS	4	办公外网	AP 接入交换机（PoE 供电）	10.10.50.2, 10.10.50.3, 10.10.50.4, 10.10.50.5（管理VLAN 50）

设备名称	型号	数量	所属网络	角色	IP地址/管理地址
Acc-Ext-AP	Cisco 3702i	80	办公外网	无线网络接入点	10.10.50.6 ~ 10.10.50.85（管理VLAN 50，每AP一个）
Switch-Ext-Server	Cisco 2960-24TT	1	办公外网	服务器接入交换机	10.10.40.56（管理VLAN 40）
Server-Ext-FW	ASA 5506	1	办公外网	服务器及出口防火墙	10.10.40.47（管理VLAN 40）
Switch-Ext-Router	Cisco 2960-24TT	1	办公外网	路由器接入交换机	10.10.40.57（管理VLAN 40）
Ext-Router	Cisco 2911	1	办公外网	互联网出口路由器	10.10.40.48（管理VLAN 40）
Ext-Syslog	Server-PT	1	办公外网	上网行为审计	10.10.40.58（管理VLAN 40）
Switch-Out	Cisco 2960-24TT			外网交换机	20.20.20.1
PC-Out				外网ip	20.20.20.2
PC-MNG-Ext			办公外网		10.10.60.1

设备名称	型号	数量	所属网络	角色	IP地址/管理地址
Finance-Server	Server-PT	1	办公外网	财务行政服务器	10.10.230.2（业务VLAN 230）
Core-Dev-1/2	Cisco 3650-24PS	2	设备专网	设备专网核心	10.10.40.49 10.10.40.50（管理VLAN 40）
Dist-Dev-1/2	Cisco 3560-24PS	2	设备专网	设备汇聚交换机	10.10.40.51（管理VLAN 40）10.10.40.52
Dist-Dev-WLC-1/2	Cisco WLC-2504	2	设备专网	移动设备无线网络控制器	10.10.60.1（管理VLAN 60） 10.10.60.86
Acc-Dev-AP-Switch	Cisco 2960-24PS	4	设备专网	移动设备AP 接入交换机（PoE 供电）	10.10.60.2, 10.10.60.3, 10.10.60.4, 10.10.60.5（管理VLAN 60）
Acc-Dev-AP	Cisco 3702i	80	设备专网	移动设备无线网络接入点	10.10.60.6 ~ 10.10.60.85（管理VLAN 60，每AP一个）
Server-Dev-FW	ASA 5506	1	设备专网	服务器区防火墙	10.10.40.53（管理VLAN 40）
Server-Dev-Switch	Cisco 2960-24TT	1	设备专网	服务器接入交换机	10.10.40.54（管理VLAN 40）

设备名称	型号	数量	所属网络	角色	IP地址/管理地址
Dev-Server	Server-PT	1	设备专网	设备专网服务器	10.10.120.2（设备VLAN 120）
Syslog-Server	Server-PT	1		日志审计服务器	10.10.40.55（VLAN 40）

```
!3650 重置端口所有配置
conf t
def int g0/1
def int g0/2
show int g 0/1
show int g 0/2
```

3 各个服务器的配置

3.1 医疗内网

3.1.1 核心层链路配置

3.1.1.1 Core-Int-1

```
enable
configure terminal
```

```
hostname Core-Int-1
no ip domain-lookup


vlan 10
 name Doctor_Workstation
exit
vlan 20
 name Nurse_Workstation
exit
vlan 30
 name Medical_Server
exit
vlan 40
 name Management
exit


interface range GigabitEthernet1/0/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
exit

interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan all
 no shutdown
exit

interface Vlan40
 ip address 10.10.40.1 255.255.255.0
 no shutdown
exit

! ========== VLAN 10（医生工作站）- VRRP 配置 ==========

interface Vlan10
```

```
ip address 10.10.10.1 255.255.254.0
no shutdown
standby 10 ip 10.10.11.254
standby 10 priority 120
standby 10 preempt
exit

! ========== VLAN 20（护士工作站）- VRRP 配置 ==========
interface Vlan20
 ip address 10.10.20.1 255.255.255.0
 no shutdown
 standby 20 ip 10.10.20.254
 standby 20 priority 120
 standby 20 preempt
exit

! ========== VLAN 30（医疗内网服务器）- VRRP 配置 ==========
interface Vlan30
 ip address 10.10.30.1 255.255.255.240
 no shutdown
 standby 30 ip 10.10.30.14
 standby 30 priority 120
 standby 30 preempt
exit

! 启用三层路由（核心交换机必备）
ip routing

! dhcp vlan10 docter

service dhcp
ip dhcp pool VLAN10-Doctor
network 10.10.10.0 255.255.254.0
default-router 10.10.11.254
ip dhcp excluded-address 10.10.10.1
ip dhcp excluded-address 10.10.10.2
ip dhcp excluded-address 10.10.11.254

! dhcp vlan 20 nurse
```

```
ip dhcp pool VLAN20-Nurse
network 10.10.20.0 255.255.255.0
default-router 10.10.20.254

ip dhcp excluded-address 10.10.20.1
ip dhcp excluded-address 10.10.20.2
ip dhcp excluded-address 10.10.20.254



! 保存配置
exit
write memory
```

3.1.1.2 Core-Int-2

```
enable
configure terminal

hostname Core-Int-2
no ip domain-lookup

! 管理VLAN
vlan 10
 name Doctor_Workstation
exit
vlan 20
 name Nurse_Workstation
exit
vlan 30
 name Medical_Server
exit
vlan 40
 name Management
exit
```

```
! 双核心互联 - 链路聚合
interface range GigabitEthernet1/0/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
exit

interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 no shutdown
exit

! 管理地址
interface Vlan40
 ip address 10.10.40.2 255.255.255.0
 no shutdown
exit

! ========== VLAN 10 - VRRP 配置（备核心） ==========
interface Vlan10
 ip address 10.10.10.2 255.255.254.0
 no shutdown
 standby 10 ip 10.10.11.254
 standby 10 priority 100
 standby 10 preempt
exit

! ========== VLAN 20 - VRRP 配置（备核心） ==========
interface Vlan20
 ip address 10.10.20.2 255.255.255.0
 no shutdown
 standby 20 ip 10.10.20.254
 standby 20 priority 100
 standby 20 preempt
exit

! ========== VLAN 30 - VRRP 配置（备核心） ==========
interface Vlan30
```

```
 ip address 10.10.30.2 255.255.255.240
 no shutdown
 standby 30 ip 10.10.30.14
 standby 30 priority 100
 standby 30 preempt
exit

! 启用三层路由
ip routing

! dhcp

ip dhcp pool VLAN10-Doctor-Backup
 network 10.10.10.0 255.255.254.0
 default-router 10.10.11.254
 lease 7 0 0

ip dhcp excluded-address 10.10.10.1
ip dhcp excluded-address 10.10.10.2
ip dhcp excluded-address 10.10.11.254

ip dhcp pool VLAN20-Nurse-Backup
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.254
 lease 7 0 0

ip dhcp excluded-address 10.10.20.1
ip dhcp excluded-address 10.10.20.2
ip dhcp excluded-address 10.10.20.254

interface Vlan10
 ip helper-address 10.10.10.1
exit
interface Vlan20
 ip helper-address 10.10.20.1
exit


service dhcp
! 医生工作站备用地址池
```

```
ip dhcp pool VLAN10-Doctor-Backup
 network 10.10.10.0 255.255.254.0
 default-router 10.10.11.254
exit

ip dhcp excluded-address 10.10.10.1
ip dhcp excluded-address 10.10.10.2
ip dhcp excluded-address 10.10.11.254

! 护士工作站备用地址池
ip dhcp pool VLAN20-Nurse-Backup
 network 10.10.20.0 255.255.255.0
 default-router 10.10.20.254
exit

ip dhcp excluded-address 10.10.20.1
ip dhcp excluded-address 10.10.20.2
ip dhcp excluded-address 10.10.20.254

! DHCP中继
interface Vlan10
 ip helper-address 10.10.10.1
exit
interface Vlan20
 ip helper-address 10.10.20.1
exit

! 保存配置
exit
write memory
```

3.1.1.3 配置验证

验证vrrp 配置

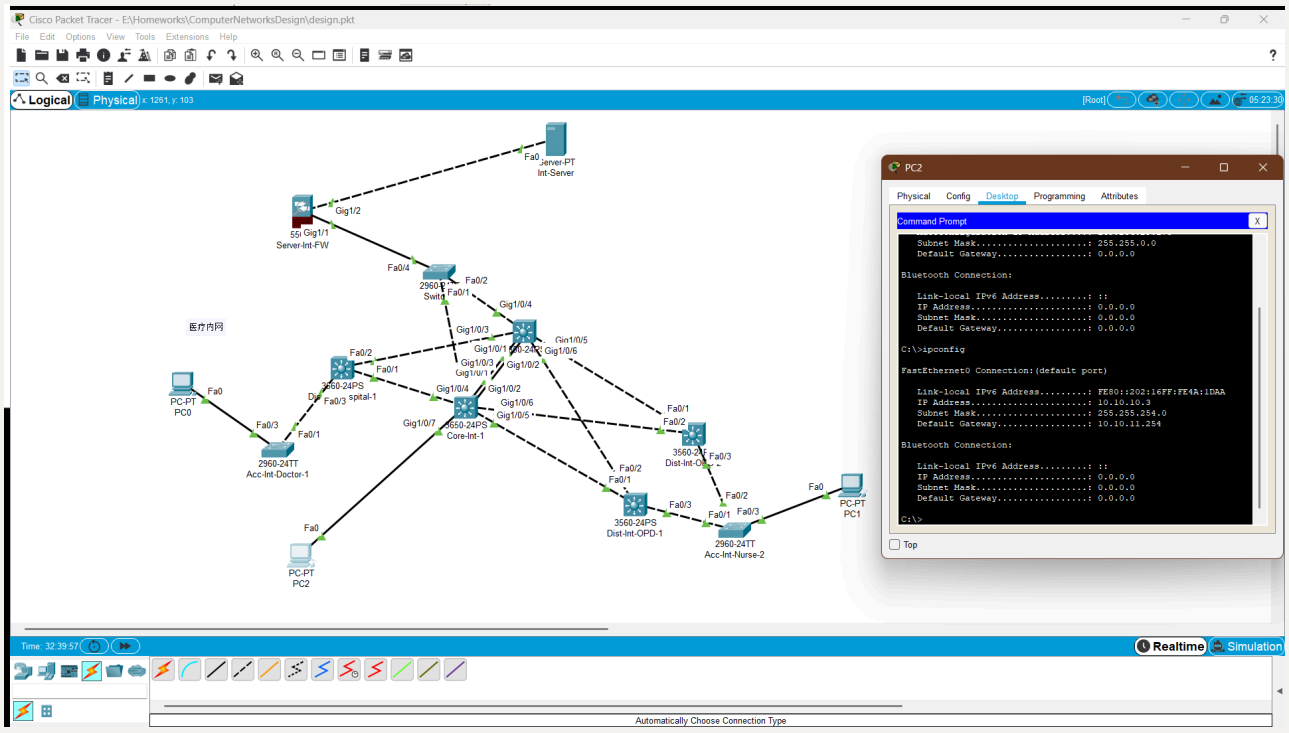
```
!Core Int 1
Core-Int-1#show standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby
Virtual IP
Vl10        10   120 P Active   local           10.10.10.2
 10.10.11.254
Vl20        20   120 P Active   local           10.10.20.2
 10.10.20.254
Vl30        30   120 P Active   local           10.10.30.2
 10.10.30.14
Vl40        40   120 P Active   local           10.10.40.2
 10.10.40.254
Core-Int-1#
```

验证dhcp设置

```
Core-Int-1#show ip dhcp pool

Pool VLAN10-Doctor :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 510
 Leased addresses               : 2
 Excluded addresses             : 6
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range
Leased/Excluded/Total
 10.10.10.1           10.10.10.1       - 10.10.11.254      2    / 6
    / 510

Pool VLAN20-Nurse :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 0
```

```
 Excluded addresses              : 6
 Pending event                   : none

 1 subnet is currently in the pool
 Current index        IP address range
Leased/Excluded/Total
 10.10.20.1           10.10.20.1       - 10.10.20.254      0    / 6
    / 254
Core-Int-1#
```

```
! 使用1/0/7接口测试
int g 1/0/7
sw mode acc
sw acc vlan 10
no sh
```



可以看到，图片中的pc2 成功获取到了dhcp分配的地址。

3.1.2 门诊汇聚层

3.1.2.1 Core-Int-1 补充配置

```
enable
configure terminal

! ========== 1. 对接 Dist-Int-OPD-1 的接口（G1/0/3） ==========
interface GigabitEthernet1/0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 2. 对接 Dist-Int-OPD-2 的接口（G1/0/4） ==========
interface GigabitEthernet1/0/4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 3. 配置管理VLAN 40的虚拟网关（HSRP） ==========
! 核心层管理VLAN需要虚拟网关，确保汇聚层运维可达
interface Vlan40
 standby 40 ip 10.10.40.254
 ! 管理VLAN虚拟网关（汇聚层默认网关指向此地址）
 standby 40 priority 120
 ! Core-Int-1作为主网关
 standby 40 preempt
 ! 抢占模式
exit

! ========== 4. 启用生成树（防环路，医院网络必备） ==========
spanning-tree mode rapid-pvst
! 快速生成树，秒级故障切换
spanning-tree vlan 10,40 root primary
! Core-Int-1作为VLAN10/40的主根桥
```

```
! 保存配置
exit
write memory
```

3.1.2.2 Core-Int-2 补充配置

```
enable
configure terminal

! ========== 1. 对接 Dist-Int-OPD-1 的接口（G1/0/3） ==========
interface GigabitEthernet1/0/3
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 2. 对接 Dist-Int-OPD-2 的接口（G1/0/4） ==========
interface GigabitEthernet1/0/4
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 3. 配置管理VLAN 40的虚拟网关（HSRP） ==========
interface Vlan40
 standby 40 ip 10.10.40.254
 ! 和Core-Int-1一致的虚拟网关
 standby 40 priority 100
 ! Core-Int-2作为备网关
 standby 40 preempt
 ! 抢占模式
exit

! ========== 4. 启用生成树 ==========
spanning-tree mode rapid-pvst
spanning-tree vlan 10,40 root secondary
```

```
! Core-Int-2作为备用根桥

! 保存配置
exit
write memory
```

3.1.2.3 Dist-Int-OPD-1 配置

```
enable
configure terminal

hostname Dist-Int-OPD-1
no ip domain-lookup

vlan 10
name Doctor_Workstation
exit

vlan 40
name Management
exit

interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

interface Vlan40
 ip address 10.10.40.5 255.255.255.0
 no shutdown
```

```
exit

ip default-gateway 10.10.40.254

interface range FastEthernet0/3 - 24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

spanning-tree mode rapid-pvst
exit
write memory
```

3.1.2.4 Dist-Int-OPD-2 配置

```
enable
configure terminal

hostname Dist-Int-OPD-2
no ip domain-lookup

vlan 10
name Doctor_Workstation
exit

vlan 40
name Management
exit

interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

interface FastEthernet0/2
```

```
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

interface Vlan40
 ip address 10.10.40.6 255.255.255.0
 no shutdown
exit

ip default-gateway 10.10.40.254

interface range FastEthernet0/3 - 24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

spanning-tree mode rapid-pvst
exit
write memory
```

3.1.2.5 配置验证

```
! ==============Dist-Int-OPD-1 ping 核心================

Dist-Int-OPD-1#ping 10.10.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/9/30
ms
```

```
Dist-Int-OPD-1#ping 10.10.40.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/5/25
ms




Dist-Int-OPD-1#show int tr
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1
Fa0/4       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,40
Fa0/2       10,40
Fa0/3       10,40
Fa0/4       10,40

Port        Vlans allowed and active in management domain
Fa0/1       10,40
Fa0/2       10,40
Fa0/3       10,40
Fa0/4       10,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       10,40
Fa0/2       10
Fa0/3       10,40
Fa0/4       10,40

Dist-Int-OPD-1#show spanning-tree vlan 10
VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
```

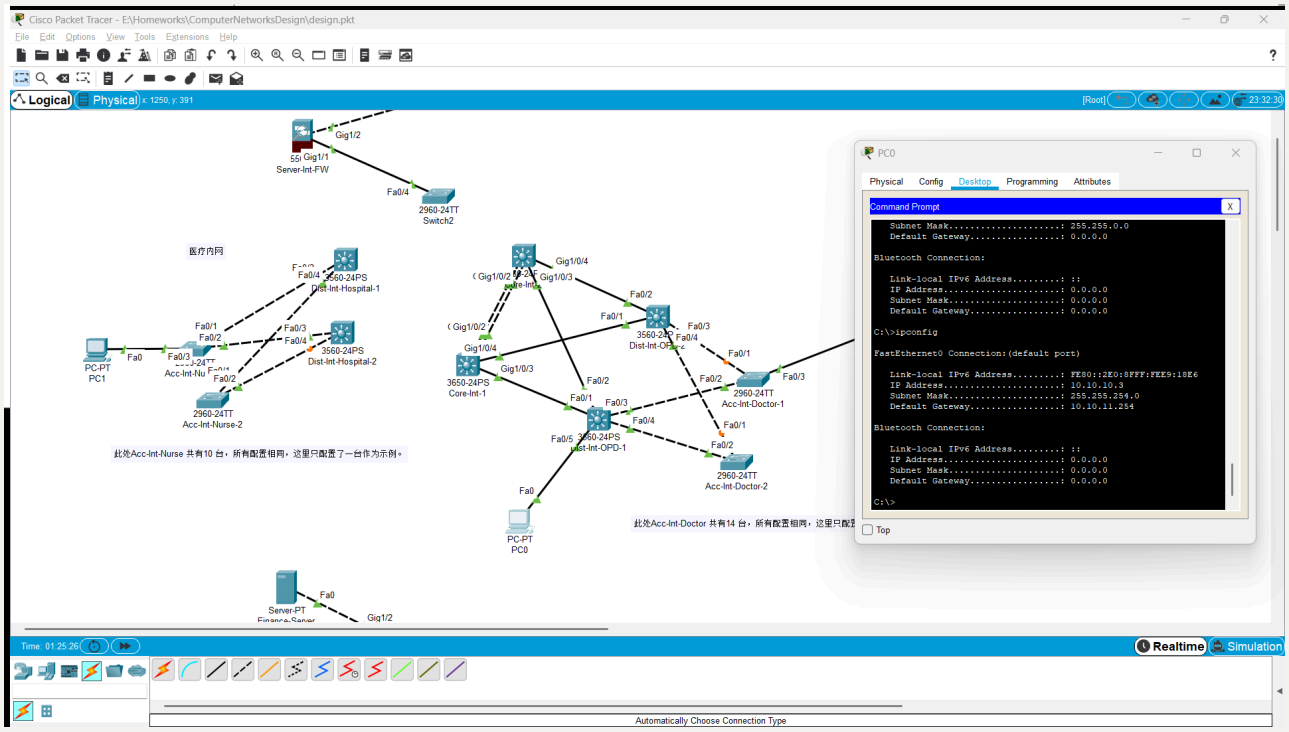
```
             Address     0001.C93B.4029
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0001.C960.EC57
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------
----------
Fa0/3            Desg FWD 19        128.3    P2p
Fa0/2            Root FWD 19        128.2    P2p
Fa0/1            Root FWD 19        128.1    P2p
Fa0/4            Desg FWD 19        128.4    P2p

Dist-Int-OPD-1#show spanning-tree vlan 40
VLAN0040
  Spanning tree enabled protocol rstp
  Root ID    Priority    24616
             Address     0001.C93B.4029
             Cost        19
             Port        1(FastEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec

  Bridge ID  Priority    28712  (priority 28672 sys-id-ext 40)
             Address     0001.C960.EC57
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------
----------
Fa0/3            Desg FWD 19        128.3    P2p
```

```
Fa0/2            Altn BLK 19        128.2    P2p
Fa0/1            Root FWD 19        128.1    P2p
Fa0/4            Desg FWD 19        128.4    P2p

Dist-Int-OPD-1#



Dist-Int-OPD-1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed
Type
Fa0/1                        connected    1          auto    auto
 10/100BaseTX
Fa0/2                        connected    1          auto    auto
 10/100BaseTX
Fa0/3                        connected    1          auto    auto
 10/100BaseTX
Fa0/4                        connected    1          auto    auto
 10/100BaseTX
Fa0/5                        notconnect   1          auto    auto
 10/100BaseTX
Fa0/6                        notconnect   1          auto    auto
 10/100BaseTX
Fa0/7                        notconnect   1          auto    auto
 10/100BaseTX
Fa0/8                        notconnect   1          auto    auto
 10/100BaseTX
Fa0/9                        notconnect   1          auto    auto
 10/100BaseTX
Fa0/10                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/11                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/12                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/13                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/14                       notconnect   1          auto    auto
 10/100BaseTX
```

```
Fa0/15                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/16                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/17                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/18                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/19                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/20                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/21                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/22                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/23                       notconnect   1          auto    auto
 10/100BaseTX
Fa0/24                       notconnect   1          auto    auto
 10/100BaseTX
Gig0/1                       notconnect   1          auto    auto
 10/100BaseTX
Gig0/2                       notconnect   1          auto    auto
 10/100BaseTX

Dist-Int-OPD-1#
Dist-Int-OPD-1#



Dist-Int-OPD-1#ping 10.10.11.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.11.254, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/18/37
ms

Dist-Int-OPD-1#
```

验证dhcp

```
!Dist-Int-OPD-1
conf t
int fa 0/5
sw mode acc
sw acc vlan 10
no sh
exit
```



3.1.3 门诊接入层

3.1.3.1 Acc-Int-Doctor-1

```
enable
configure terminal

! 基础配置
hostname Acc-Int-Doctor-1
no ip domain-lookup
```

```
! 创建业务VLAN和管理VLAN
vlan 10
 name Doctor_Workstation
exit
vlan 40
 name Management
exit

! ========== 上联 Dist-Int-OPD-1（Fa0/3） ==========
interface FastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 上联 Dist-Int-OPD-2（Fa0/3） ==========
interface FastEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 10,40
 no shutdown
exit

! ========== 管理VLAN配置（运维用） ==========
interface Vlan40
 ip address 10.10.40.7 255.255.255.0
 no shutdown
exit

! 管理网关指向核心虚拟网关
ip default-gateway 10.10.40.254

! ========== 下联医生工作站PC（所有终端接口） ==========
interface range FastEthernet0/3 - 24
 switchport mode access
 switchport access vlan 10
 no shutdown
exit

! ========== 生成树（防环路，冗余必备） ==========
```

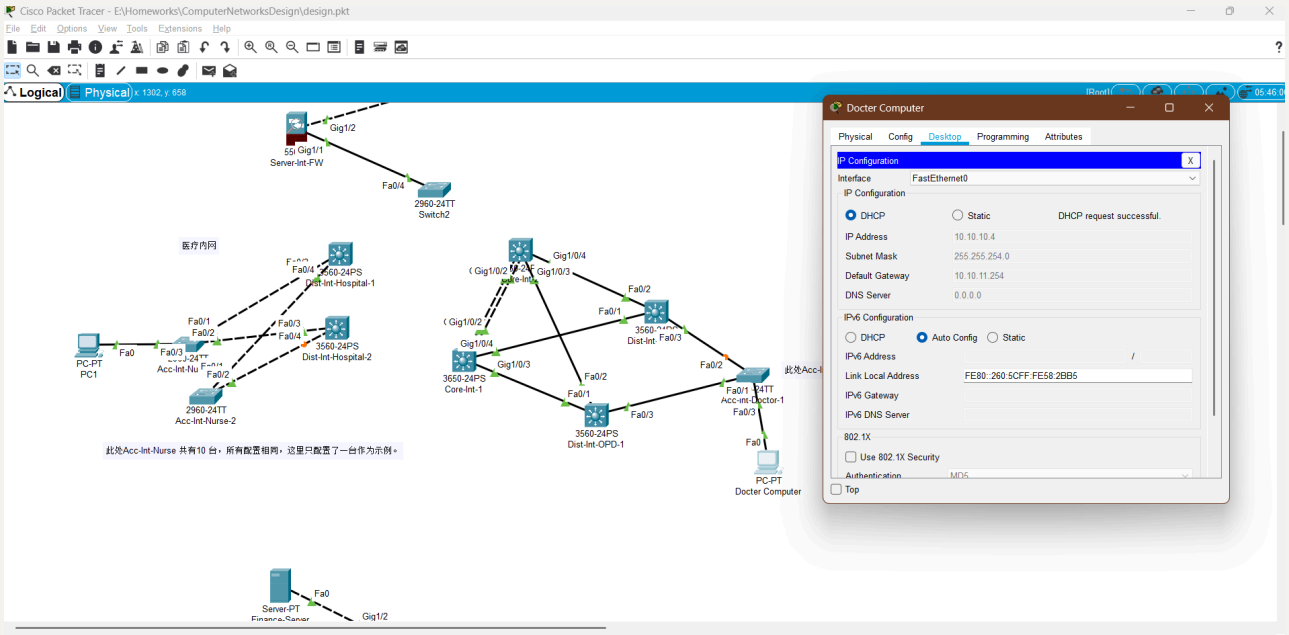
```
spanning-tree mode rapid-pvst

! 保存配置
exit
write memory
```

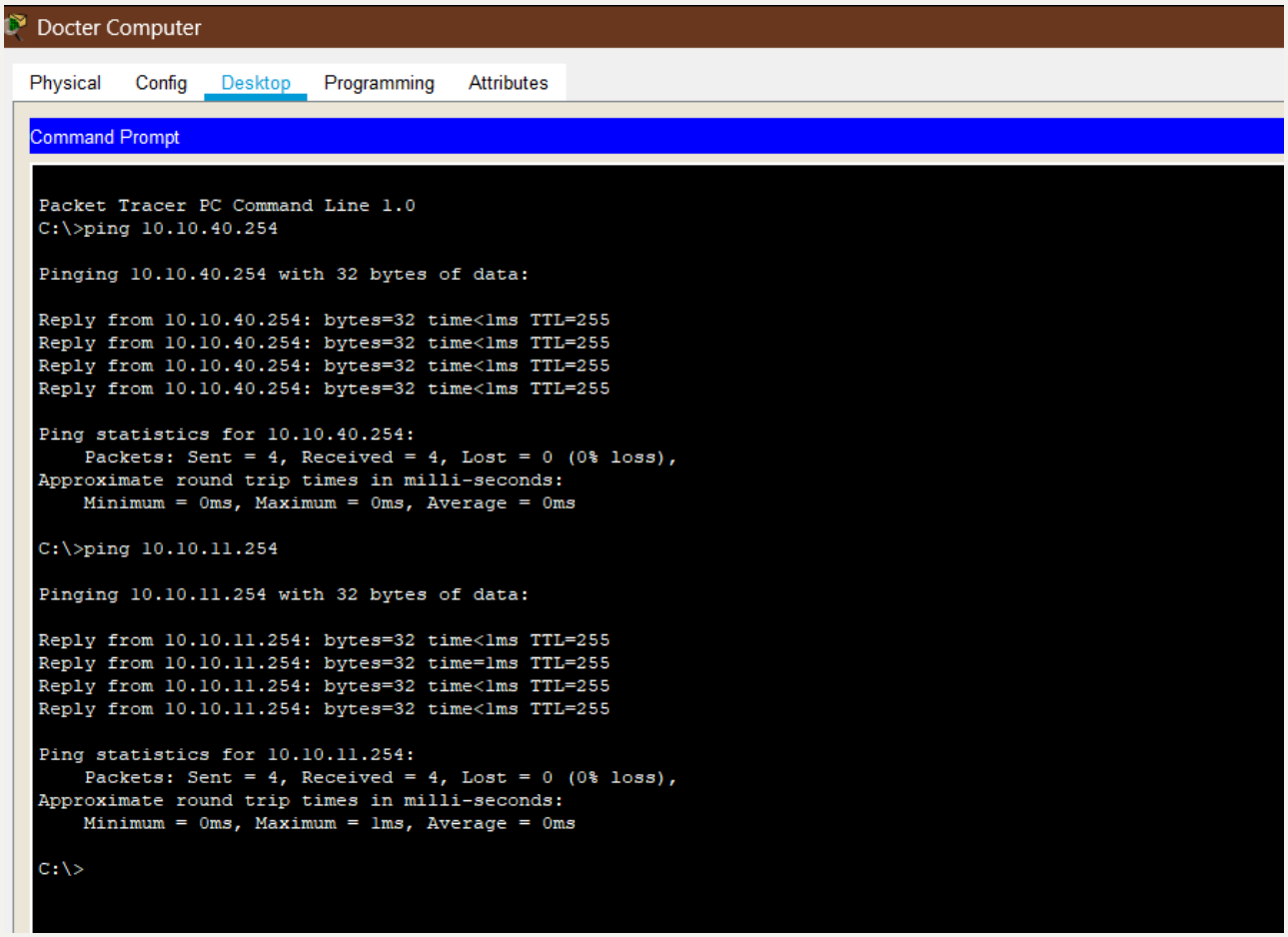
3.1.3.2 配置验证

```
Acc-Int-Doctor-1#show int trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       10,40
Fa0/2       10,40

Port        Vlans allowed and active in management domain
Fa0/1       10,40
Fa0/2       10,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       40
Fa0/2       none

Acc-Int-Doctor-1#
```

Fa 0/2 是防止环路的行为，none是正常的结果

dhcp ip获取正常


Cisco Packet Tracer - E:\Homeworks\ComputerNetworksDesign\design.pkt
Logical
Physical
Server-Int-FW
2960-24TT
Switch2
医疗内网
Dist-Int-Hospital-1
Dist-Int-Hospital-2
PC-PT
PC1
Acc-Int-Nu
Acc-Int-Nurse-2
Core-Int-1
Dist-Int-OPD-1
Acc-Int-Doctor-1
PC-PT
Docter Computer
此处Acc-Int-Nurse 共有10 台，所有配置相同，这里只配置了一台作为示例。
Docter Computer
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
DHCP
Static
DHCP request successful.
IP Address
10.10.10.4
Subnet Mask
255.255.254.0
Default Gateway
10.10.11.254
DNS Server
0.0.0.0
IPv6 Configuration
Auto Config
IPv6 Address
Link Local Address
FE80::260:5CFF:FE58:2BB5
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Top


Docter Computer
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 10.10.40.254
Pinging 10.10.40.254 with 32 bytes of data:
Reply from 10.10.40.254: bytes=32 time<1ms TTL=255
Reply from 10.10.40.254: bytes=32 time<1ms TTL=255
Reply from 10.10.40.254: bytes=32 time<1ms TTL=255
Reply from 10.10.40.254: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.40.254:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 10.10.11.254
Pinging 10.10.11.254 with 32 bytes of data:
Reply from 10.10.11.254: bytes=32 time<1ms TTL=255
Reply from 10.10.11.254: bytes=32 time=1ms TTL=255
Reply from 10.10.11.254: bytes=32 time<1ms TTL=255
Reply from 10.10.11.254: bytes=32 time<1ms TTL=255
Ping statistics for 10.10.11.254:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

网关均能ping通

3.1.4 护士工作站汇聚层

3.1.4.1 Core-Int-1 补充配置（对接护士汇聚层）

```
enable
configure terminal

! ========== 1. 对接 Dist-Int-Hospital-1 的接口（G1/0/5） ==========
interface GigabitEthernet1/0/5
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 2. 对接 Dist-Int-Hospital-2 的接口（G1/0/6） ==========
interface GigabitEthernet1/0/6
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 3. 生成树强化配置（护士VLAN 20） ==========
spanning-tree vlan 20 root primary
! Core-Int-1作为VLAN20主根桥

! 保存配置
exit
write memory
```

3.1.4.2 Core-Int-2 补充配置（对接护士汇聚层）

```
enable
configure terminal

! ========== 1. 对接 Dist-Int-Hospital-1 的接口（G1/0/5） ==========
interface GigabitEthernet1/0/5
```

```
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 2. 对接 Dist-Int-Hospital-2 的接口（G1/0/6） ==========
interface GigabitEthernet1/0/6
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 3. 生成树强化配置（护士VLAN 20） ==========
spanning-tree vlan 20 root secondary
! Core-Int-2作为VLAN20备用根桥

! 保存配置
exit
write memory
```

3.1.4.3 Dist-Int-Hospital-1 配置（护士汇聚层1）

```
enable
configure terminal

hostname Dist-Int-Hospital-1
no ip domain-lookup

vlan 20
name Nurse_Workstation
exit

vlan 40
name Management
exit

interface FastEthernet0/1
```

```
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

interface Vlan40
 ip address 10.10.40.3 255.255.255.0
 no shutdown
exit

ip default-gateway 10.10.40.254

interface range FastEthernet0/3 - 24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

spanning-tree mode rapid-pvst
exit
write memory
```

3.1.4.4 Dist-Int-Hospital-2 配置（护士汇聚层2）

```
enable
configure terminal

hostname Dist-Int-Hospital-2
no ip domain-lookup
```

```
vlan 20
name Nurse_Workstation
exit

vlan 40
name Management
exit

interface FastEthernet0/1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

interface FastEthernet0/2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

interface Vlan40
 ip address 10.10.40.4 255.255.255.0
 no shutdown
exit

ip default-gateway 10.10.40.254

interface range FastEthernet0/3 - 24
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

spanning-tree mode rapid-pvst
exit
write memory
```

3.1.4.5 配置验证

```
Core-Int-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Po1         on           802.1q         trunking      1
Gig1/0/3    on           802.1q         trunking      1
Gig1/0/4    on           802.1q         trunking      1
Gig1/0/5    on           802.1q         trunking      1
Gig1/0/6    on           802.1q         trunking      1

Port        Vlans allowed on trunk
Po1         1-1005
Gig1/0/3    10,40
Gig1/0/4    10,40
Gig1/0/5    20,40
Gig1/0/6    20,40

Port        Vlans allowed and active in management domain
Po1         1,10,20,30,40
Gig1/0/3    10,40
Gig1/0/4    10,40
Gig1/0/5    20,40
Gig1/0/6    20,40

Port        Vlans in spanning tree forwarding state and not pruned
Po1         1,10,20,30,40
Gig1/0/3    10,40
Gig1/0/4    10,40
Gig1/0/5    20,40
Gig1/0/6    20,40

Core-Int-1#show spanning-tree vlan 20
VLAN0020
  Spanning tree enabled protocol rstp
  Root ID    Priority    24596
             Address     0001.C93B.4029
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec
```

```
  Bridge ID  Priority    24596  (priority 24576 sys-id-ext 20)
             Address     0001.C93B.4029
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15
sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ----------------------
----------
Gi1/0/4          Desg FWD 19        128.4    P2p
Gi1/0/3          Desg FWD 19        128.3    P2p
Gi1/0/5          Desg FWD 19        128.5    P2p
Gi1/0/6          Desg FWD 19        128.6    P2p
Po1              Desg FWD 3         128.29   Shr
```

```
Dist-Int-Hospital-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1
Fa0/3       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       20,40
Fa0/2       20,40
Fa0/3       20,40

Port        Vlans allowed and active in management domain
Fa0/1       20,40
Fa0/2       20,40
Fa0/3       20,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,40
Fa0/2       none
Fa0/3       20,40

Dist-Int-Hospital-1#ping 10.10.40.254

Type escape sequence to abort.
```

```
Sending 5, 100-byte ICMP Echos to 10.10.40.254, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0
ms

Dist-Int-Hospital-1#ping 10.10.20.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.254, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0
ms

Dist-Int-Hospital-1#ping 10.10.40.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1
ms
```

3.1.5 护士工作站接入层

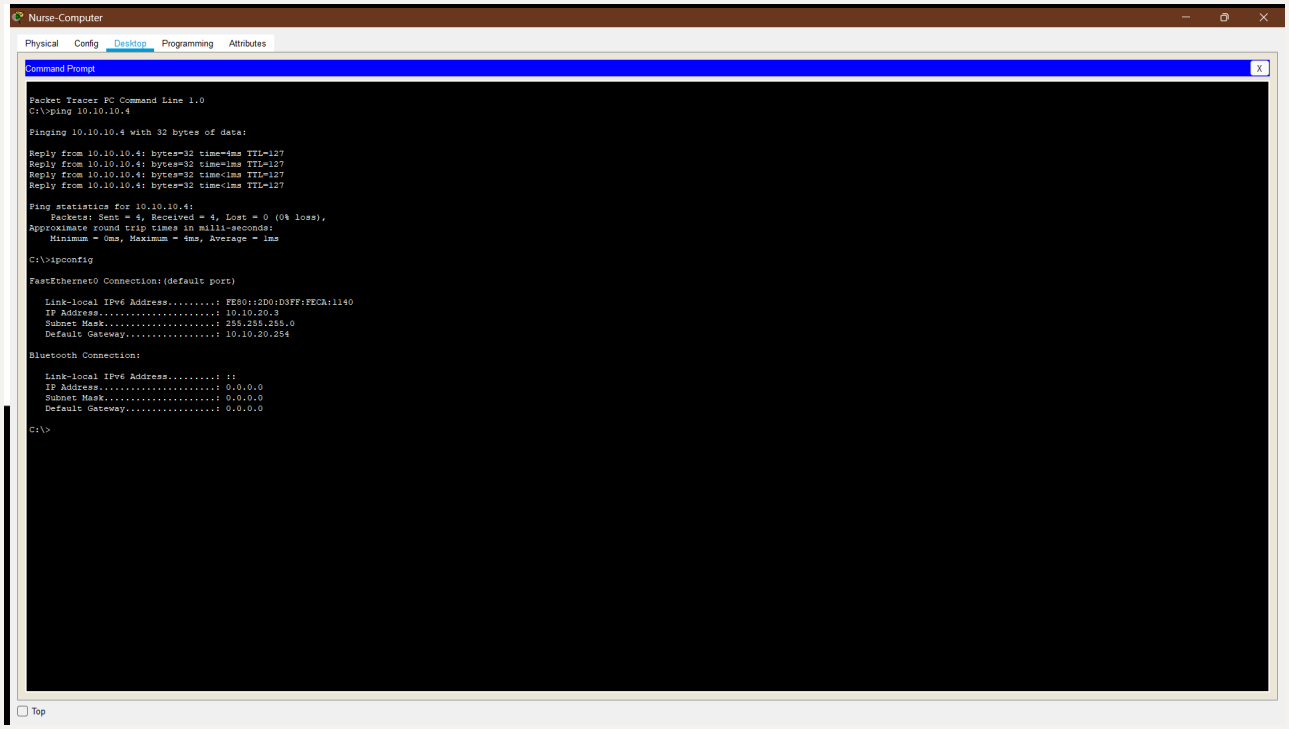
3.1.5.1 Acc-Int-Nurse-1

```
enable
configure terminal

! 基础配置
hostname Acc-Int-Nurse-1
no ip domain-lookup

! 创建业务VLAN和管理VLAN
vlan 20
 name Nurse_Workstation
exit
```

```
vlan 40
 name Management
exit

! ========== 上联 Dist-Int-Hospital-1（Fa0/3） ==========
interface FastEthernet0/1
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 上联 Dist-Int-Hospital-2（Fa0/3） ==========
interface FastEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 20,40
 no shutdown
exit

! ========== 管理VLAN配置（运维用） ==========
interface Vlan40
 ip address 10.10.40.21 255.255.255.0
 no shutdown
exit

! 管理网关指向核心虚拟网关
ip default-gateway 10.10.40.254

! ========== 下联护士工作站PC（所有终端接口） ==========
interface range FastEthernet0/3 - 24
 switchport mode access
 switchport access vlan 20
 no shutdown
exit

! ========== 生成树（防环路，冗余必备） ==========
spanning-tree mode rapid-pvst

! 保存配置
exit
write memory
```

3.1.5.2 快速验证命令（配置后执行）

```
Acc-Int-Nurse-1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1
Fa0/2       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       20,40
Fa0/2       20,40

Port        Vlans allowed and active in management domain
Fa0/1       20,40
Fa0/2       20,40

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       20,40
Fa0/2       none

Acc-Int-Nurse-1#show ip interface brief | include Vlan40
Vlan40                  10.10.40.21     YES manual up
     up


Acc-Int-Nurse-1#show interfaces FastEthernet0/3 switchport |
include Access Mode VLAN
Access Mode VLAN: 20 (Nurse_Workstation)
```



护士电脑能正确入网，dhcp获取ip地址，并能正确地和医生通讯

3.1.6 服务器

3.1.6.1 Core-Int-1 补充配置

```
enable
configure terminal

! 对接防火墙GigabitEthernet1/1的接口（G1/0/7）
interface GigabitEthernet1/0/7
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 30,40
 no shutdown
exit

! VLAN30主根桥（服务器网段生成树主节点）
spanning-tree vlan 30 root primary
```

```
! 保存配置
exit
write memory
```

3.1.6.2 Core-Int-2 补充配置

```
enable
configure terminal

! 对接防火墙GigabitEthernet1/2的接口（G1/0/7）
interface GigabitEthernet1/0/7
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan 30,40
 no shutdown
exit

! VLAN30备用根桥（核心故障时自动切换）
spanning-tree vlan 30 root secondary

! 保存配置

exit
write memory
```

3.1.6.3 Server-Int-Switch

```
! 进入特权模式
enable
configure terminal

hostname Server-Int-Switch
no ip domain-lookup

! ===================== 2. VLAN配置（匹配医疗内网规划）
=====================
! 医疗服务器业务VLAN
```

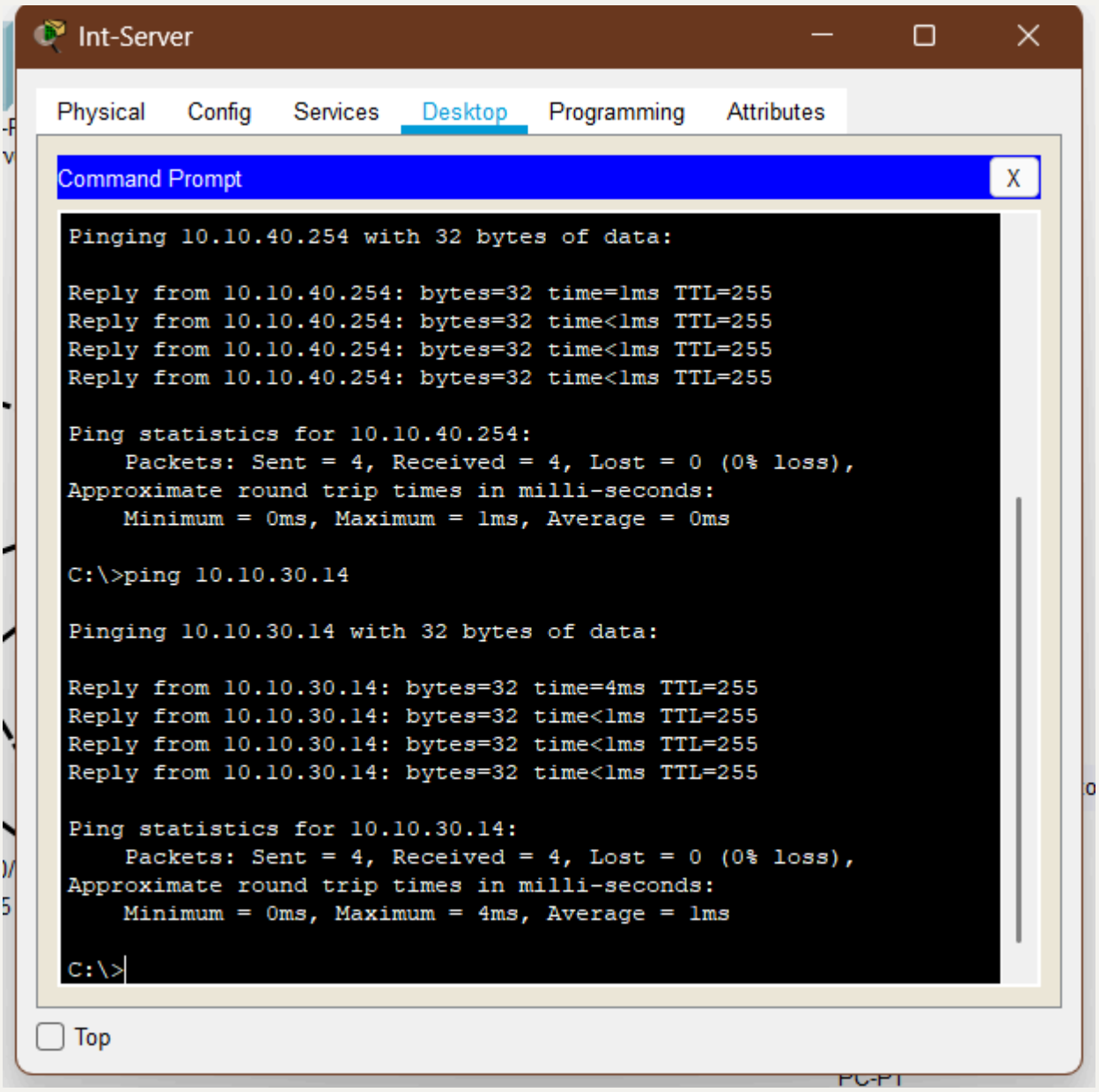
```
vlan 30
 name Medical_Server
exit

! 设备管理VLAN（与核心/防火墙统一）
vlan 40
 name Management
exit

! ===================== 3. 上联核心交换机接口配置（冗余连通）
=====================
! Fa0/1 对接 Core-Int-1 的 G1/0/7
interface FastEthernet0/1
 switchport mode trunk

 switchport trunk allowed vlan 30,40
 no shutdown
exit

! Fa0/2 对接 Core-Int-2 的 G1/0/7（冗余备份）
interface FastEthernet0/2
 switchport mode trunk
 switchport trunk allowed vlan 30,40
 no shutdown
exit

! ===================== 4. 服务器接入端口配置（核心接入）
=====================
interface range FastEthernet0/3 - 24
 switchport mode access
 switchport access vlan 30
 spanning-tree portfast
 no shutdown
exit

! ===================== 5. 管理VLAN配置（保障设备运维）
=====================
interface Vlan40
 ip address 10.10.40.32 255.255.255.0
 no shutdown
```

```
exit

! 默认网关（指向核心层VRRP虚拟网关，保障冗余）
ip default-gateway 10.10.40.254

! ===================== 6. 生成树配置（防环路，保障冗余）
=====================
spanning-tree mode rapid-pvst
spanning-tree vlan 30,40 priority 24576

! ===================== 7. 保存配置（防止丢失） =====================
exit
write memory
```

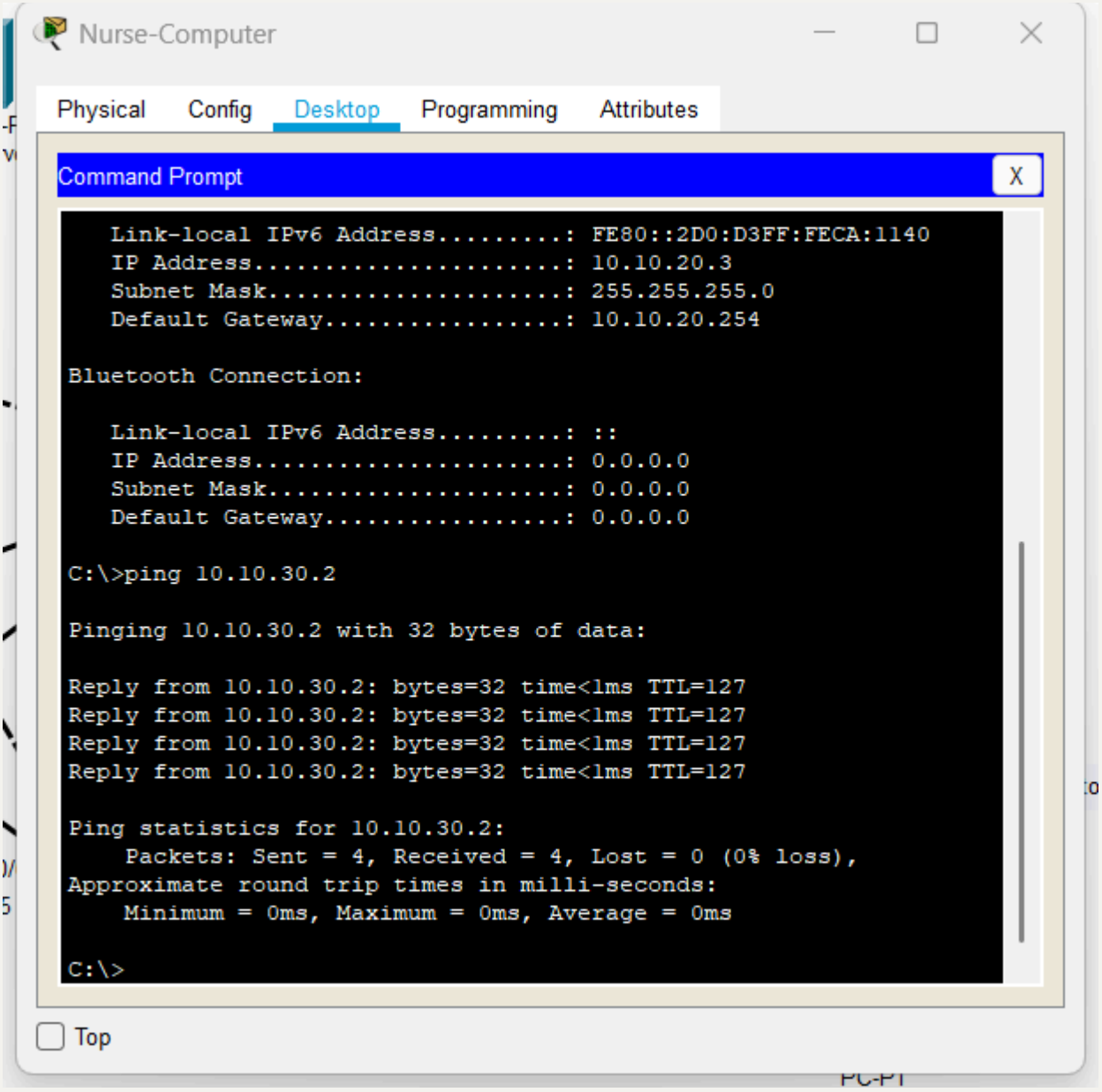
3.1.6.3 测试配置

```
Server-Int-Switch#ping 10.10.40.254

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.40.254, timeout is 2
seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1
ms

Server-Int-Switch#
```

服务器ping网关



护士电脑ping服务器



3.1.7 核心服务器配置acl

3.1.7.1 Core-Int-1

```
enable
configure terminal

! ========== 1. 定义极简ACL（放行所有IP流量） ==========
ip access-list extended MEDICAL_SERVER_ACL
 permit ip any any
exit
```

```
! ========== 2. 应用到服务器VLAN三层接口（Vlan30） ==========
interface Vlan30
 ip access-group MEDICAL_SERVER_ACL in
exit

! ========== 3. 保存配置 ==========
exit
write memory
```

3.1.7.2 Core-Int-2

```
enable
configure terminal

! ========== 1. 同步极简ACL配置 ==========
ip access-list extended MEDICAL_SERVER_ACL
 permit ip any any
exit

! ========== 2. 应用到Vlan30接口 ==========
interface Vlan30
 ip access-group MEDICAL_SERVER_ACL in
exit

! ========== 3. 保存配置 ==========
exit
write memory
```

3.2 政务外网

3.2.1 核心层配置

3.2.1.1 Core-Ext-1

```
enable
configure terminal

hostname Core-Ext-1
no ip domain-lookup

! 创建办公外网相关VLAN
vlan 210
 name Admin_Office
exit
vlan 220
 name Ext_Router
exit
vlan 230
 name Finance_Server
exit
vlan 240
 name Guest_Wireless
exit
vlan 40
 name Management
exit
vlan 50
 name AP_Management_Ext
exit

! 双核心互联 - 链路聚合配置
interface range GigabitEthernet1/0/1 - 2
 switchport trunk encapsulation dot1q
 switchport mode trunk
 channel-group 1 mode active
 no shutdown
exit

interface Port-channel1
 switchport trunk encapsulation dot1q
 switchport mode trunk
 switchport trunk allowed vlan all
 no shutdown
```

```
exit

! 管理VLAN配置
interface Vlan40
 ip address 10.10.40.33 255.255.255.0
 no shutdown
 standby 40 ip 10.10.40.254
 standby 40 priority 120
 standby 40 preempt
exit

! AP管理VLAN配置（办公外网AP）
interface Vlan50
 ip address 10.10.50.1 255.255.255.0
 no shutdown
 standby 50 ip 10.10.50.254
 standby 50 priority 120
 standby 50 preempt
exit

! ========== VLAN 210（行政办公电脑）- VRRP 配置 ==========
interface Vlan210
ip address 10.10.210.1 255.255.255.0
no shutdown
standby 210 ip 10.10.210.254
standby 210 priority 120
standby 210 preempt
exit

! ========== VLAN 220（外网路由器）- VRRP 配置 ==========
interface Vlan220
 ip address 10.10.220.1 255.255.255.240
 no shutdown
 standby 220 ip 10.10.220.14
 standby 220 priority 120
 standby 220 preempt
exit

! ========== VLAN 230（财务行政服务器）- VRRP 配置 ==========
interface Vlan230
```

```
 ip address 10.10.230.1 255.255.255.240
 no shutdown
 standby 230 ip 10.10.230.14
 standby 230 priority 120
 standby 230 preempt
exit

! ========== VLAN 240（客户病房无线网络）- VRRP 配置 ==========
interface Vlan240
 ip address 10.20.0.1 255.255.240.0
 no shutdown
 standby 240 ip 10.20.15.254
 standby 240 priority 120
 standby 240 preempt
exit

! 启用三层路由
ip routing

! DHCP配置 - VLAN210行政办公
service dhcp
ip dhcp pool VLAN210-Admin
network 10.10.210.0 255.255.255.0
default-router 10.10.210.254
ip dhcp excluded-address 10.10.210.1
ip dhcp excluded-address 10.10.210.2
ip dhcp excluded-address 10.10.210.254

! DHCP配置 - VLAN240访客无线
ip dhcp pool VLAN240-Guest
network 10.20.0.0 255.255.240.0
default-router 10.20.15.254
lease 1 0 0  ! 访客地址租期1天
ip dhcp excluded-address 10.20.0.1
ip dhcp excluded-address 10.20.0.2
ip dhcp excluded-address 10.20.15.254

! 生成树配置
spanning-tree mode rapid-pvst
spanning-tree vlan 210,220,230,240,50 root primary
```

```
! 保存配置
exit
write memory
```